MESA SCHOOL OF BUSINESS

Data Foundations · Pre-Read

Session 8: Groww

Big Mastery Question: Build the founder a list that updates itself — every leaking funnel step, refreshed every Monday without you.

Before you come to class

Read this once, slowly. Every formula so far has lived in ONE cell. This session introduces formulas that spill across many cells automatically, and one function (QUERY) that behaves like a small database inside a single cell. This is the most powerful session yet — take it slow. Every table below shows real column letters and row numbers, as always.

The company: Groww is an investing app — users sign up, complete KYC (identity verification), and eventually place their first trade. Every step a user takes logs one event row. Your job this session: find exactly where users are getting stuck in that funnel, and build a list that updates itself automatically — no more manually re-running the same report every week.

0. The Dataset You'll Be Working With

Groww's funnel events table has 8 columns (event_id, user_id, event_date, event_name, channel, device, state, referral_code). Here's a 6-column, 6-row slice:

	A	B	C	D	E	F
1	event_id	user_id	event_date	event_name	channel	device
2	GE001	U001	2026-07-01	signup	Paid Social	Android
3	GE002	U001	2026-07-02	kyc_started	Paid Social	Android
4	GE003	U002	2026-07-03	signup	Organic	iOS
5	GE004	U002	2026-07-04	kyc_started	Organic	iOS
6	GE005	U002	2026-07-05	kyc_done	Organic	iOS
7	GE006	U003	2026-07-06	signup	Referral	Android

Column A = event_id, B = user_id, C = event_date, D = event_name, E = channel, F = device. (G = state, H = referral_code, not shown.) Notice each user leaves a TRAIL of rows — one per step they've completed.

1. FILTER

Real world reference:

Remember the magic sunglasses from Filter Views back in Session 1 — they hid rows temporarily, without changing the data. FILTER does something similar but more powerful: it creates a brand NEW, separate list containing only the matching rows, that updates itself automatically as your data changes.

In Groww's world:

You want a standing list of every user currently stuck at "kyc_started" — meaning they began verification but never finished it. FILTER scans event_name for that exact stage and spills out a live list of just those rows — no manual filtering, no re-running anything.

Syntax:

```
=FILTER(range, condition1, [condition2, ...])  Example: =FILTER(A2:F7, D2:D7="kyc_started")
```

See it in the table:

	K	L	M	N	O
1	event_id	user_id	event_date	event_name	channel
2	GE002	U001	2026-07-02	kyc_started	Paid Social
3	GE004	U002	2026-07-04	kyc_started	Organic

Typed ONCE in K2, this formula pulled back exactly the 2 rows that match — and if a new user gets stuck at kyc_started next week, this list grows on its own.

2. SORT

Real world reference:

This is the FORMULA version of the Sort menu action from Session 1. Instead of physically reordering rows on the sheet, SORT creates a new, reordered list from a range — leaving the original untouched.

In Groww's world:

You want your "stuck at KYC" list from FILTER to show the MOST RECENT cases first, so support calls the freshest drop-offs before they go cold. SORT can even wrap directly around a FILTER formula, sorting its results on the fly.

Syntax:

```
=SORT(range, sort_column, is_ascending)  Example:
=SORT(FILTER(A2:F7,D2:D7="kyc_started"), 3, FALSE)
```

See it in the table:

	K	L	M	N
1	event_id	user_id	event_date	event_name
2	GE004	U002	2026-07-04	kyc_started
3	GE002	U001	2026-07-02	kyc_started

Same two rows as the FILTER example, but now GE004 (04 July, more recent) appears before GE002 (02 July) — sort_column 3 refers to the 3rd column of the FILTERED result (event_date), and FALSE means descending.

3. UNIQUE

Real world reference:

Imagine a bag with several duplicate marbles in it, and you want to know every DISTINCT color present — just one of each, no repeats. UNIQUE does exactly that to a column of data.

In Groww's world:

Before building a dropdown or a report, you want to know every channel that appears in the data — without scrolling through hundreds of repeated "Paid Social" and "Organic" entries. UNIQUE gives you the distinct list instantly.

Syntax:

```
=UNIQUE(range)  Example: =UNIQUE(E2:E7)
```

See it in the table:

	E
1	channel (raw, with repeats)
2	Paid Social
3	Paid Social
4	Organic
5	Organic
6	Organic
7	Referral

=UNIQUE(E2:E7) collapses all 6 rows down to just the distinct values:

	K
1	Unique Channels
2	Paid Social
3	Organic
4	Referral

4. SORTN

Real world reference:

If SORT reorders an entire list, SORTN is SORT with a bouncer standing at the door — it sorts everything, but only lets the TOP few through. "Give me just the top 2, sorted" in a single step.

In Groww's world:

You don't want the entire event log sorted by date — you just want the 2 MOST RECENT events across the whole funnel, for a quick "what just happened" glance. SORTN sorts and trims in one move.

Syntax:

```
=SORTN(range, n, ties_mode, sort_column, is_ascending)  Example: =SORTN(A2:D7, 2, 0, 3, FALSE)
```

See it in the table:

	K	L	M	N
1	event_id	user_id	event_date	event_name

2	GE006	U003	2026-07-06	signup
3	GE005	U002	2026-07-05	kyc_done

Out of all 6 events, only the 2 most recent survive — sorted descending by event_date (column 3), exactly like SORT, but automatically trimmed to the top N.

5. SEQUENCE

Real world reference:

Like asking someone to write out the numbers 1 through 5 on a piece of paper, one under another — SEQUENCE generates a run of numbers automatically, without you typing each one by hand.

In Groww's world:

When you build a cohort or funnel-step table later this term, you'll often need a clean column of step numbers — Day 1, Day 2, Day 3... or Step 1, Step 2, Step 3. SEQUENCE generates that instantly instead of typing 1, 2, 3 by hand and hoping you don't make a typo on row 47.

Syntax:

```
=SEQUENCE(rows)  Example: =SEQUENCE(5)
```

See it in the table:

	K
1	**Funnel Step #**
2	=SEQUENCE(5) → 1
3	2
4	3
5	4
6	5

One formula, typed once in K2, spilled 5 numbers straight down — no manual typing, no risk of a duplicate or skipped number.

6. ARRAYFORMULA

Real world reference:

Remember Autofill from Session 1 — writing one formula, then dragging it down every row by hand? ARRAYFORMULA is Autofill's upgrade: write the formula ONCE, and it automatically applies itself down an entire column, with nothing to drag and nothing that breaks if new rows get added later.

In Groww's world:

You want a flag column marking every row that's a KYC step ("kyc_started" or "kyc_done") — for all 6 rows, without writing the formula 6 times or dragging a fill handle. ARRAYFORMULA wraps around an ordinary IF and makes it apply to the whole range in one shot.

Syntax:

```
=ARRAYFORMULA(formula_using_ranges)  Example:
=ARRAYFORMULA(IF((D2:D7="kyc_started")+(D2:D7="kyc_done"),"KYC Step","Other"))
```

See it in the table:

	A	D	K
1	**event_id**	**event_name**	**Funnel Stage**
2	GE001	signup	Other
3	GE002	kyc_started	KYC Step
4	GE003	signup	Other
5	GE004	kyc_started	KYC Step
6	GE005	kyc_done	KYC Step
7	GE006	signup	Other

ONE formula in K2 filled all 6 rows of column K instantly. If 500 more rows get added to the sheet tomorrow, this formula extends to cover them too — a dragged Autofill formula would NOT.

7. QUERY

Real world reference:

Imagine being able to ask your spreadsheet a question in almost plain English — "show me the count of events, grouped by channel" — and have it just answer you, instead of you building a pivot table by clicking through menus. QUERY is a tiny database language living inside a single Sheets formula.

In Groww's world:

You want a count of events per channel, as a single live formula rather than a manual pivot table — useful when you want the summary to sit right next to other formulas, or feed into a dashboard automatically.

Syntax:

```
=QUERY(range, "SELECT col, COUNT(col2) WHERE ... GROUP BY col ORDER BY ... LABEL
...", headers)  Example: =QUERY(A1:F7, "SELECT E, COUNT(A) GROUP BY E LABEL COUNT(A)
'Events'", 1)
```

See it in the table:

	K	L
1	**channel**	**Events**
2	Paid Social	2
3	Organic	3
4	Referral	1

SELECT chooses what to show (channel, and a count), GROUP BY buckets rows by channel, and LABEL renames the count column from the default "count A" to something readable — "Events". This is the exact same result a pivot table would give you, but as one portable formula.

Coming to class

You now know how to build lists that update themselves, generate sequences, and query your data like a mini database. In class, we'll build the live, self-updating funnel-leak report the founder can open every Monday without you touching it again.